AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-08-17 | Entry window 17 August 2026 | 12 records carried, 9 rejected

12 RECORDS CARRIED (9 REJECTED)	**8** SUBTOPIC CLUSTERS REPRESENTED	**10** EFFECT ESTIMATES WITH INTERVALS	**6** INSTRUMENTATION / EXPOSURE RECORDS	**2** METADATA-ONLY (LABELLED)

Signal of the day

The constituent ordering inverts against yesterday's issue: here the secondary *inorganic* ions lead and organic matter is the fraction that collapses under adjustment.

Zhao et al. (*Chin Med J*) follow **20,117 pregnancies** in Nanjing with residential $PM_{2.5}$ resolved into nitrate, sulfate, ammonium, black carbon and organic matter, and estimate spontaneous-abortion odds per IQR by logistic regression with propensity-score matching, WQS and BKMR. Total mass gives **OR 1.18 (95% CI 1.11–1.26)**; the ordering across constituents is **NH_4^+ 1.19 (1.11–1.26)**, **NO_3^- 1.18 (1.11–1.26)**, **SO_4^{2-} 1.17 (1.10–1.25)**, then **OM 1.14 (1.07–1.21)** and **BC 1.14 (1.08–1.22)**. After further adjustment for season of conception, OM (1.08, 0.99–1.18) and BC (1.07, 0.97–1.18) go null while the three inorganic ions hold. That is the exact reverse of the 16 August pairing, where Wu et al. attributed 54.1% of a prenatal joint effect to organic matter. The two are not in contradiction — different outcome, different cohort, different exposure product — but taken together they say the constituent that carries a prenatal association is not stable across studies, and the seasonal-confounding structure is doing more work than either paper resolves. *For instrumentation the implication is sharper than yesterday's*: ammonium nitrate and ammonium sulfate are the most hygroscopic fractions of the aerosol, so the constituents that survive adjustment here are precisely the ones a nephelometric low-cost sensor over-reads at high RH and precisely the ones a humidity-correction curve is fitted to remove. A network calibrated to reproduce reference *mass* is being tuned against the signal this study says matters.

What changed today

- **The greenness-modification claim in the same paper does not hold up to its own numbers.** Zhao et al. report a significant $PM_{2.5}\times$NDVI-1000 m interaction and read it as greenness attenuating risk, but the quartile series is **Q1 1.30 (1.14–1.48), Q2 1.02 (0.90–1.15), Q3 1.29 (1.14–1.46), Q4 1.11 (0.97–1.26)** — Q3 is statistically indistinguishable from Q1. That is non-monotone, not a gradient, and the interaction p-value is being asked to carry a dose–response story the strata do not support.
- **A childhood-leukaemia signal arrives on a very thin exposure contrast.** Choi et al. (*Environ Res*) report **HR 1.40 (1.04–1.90) per 5 $\mu g/m^3$** across 384,606 Korean children, but the cohort mean is **29.6 $\mu g/m^3$ with an SD of only 3.1** and exposure is assigned at district level from an ML ensemble. A 5 $\mu g/m^3$ increment is more than 1.5 SD of the available between-district spread, so the estimate is an extrapolation across most of its own range on 272 events.
- **Proteomic signatures are proposed as a replacement for the ambient concentration — and cannot be checked here.** Chen et al. (*Ecotox Environ Saf*) screen ∼2,900 UK Biobank plasma proteins against seven pollutants, build elastic-net signatures, and report they out-predict external exposure metrics for incident AF, with ANGPT2, GDF15 and PLAUR as mediators and **1.15–1.25 years of AF-free life lost** in the top-risk half. No hazard ratio or interval appears in the abstract, so the record carries *no* estimate into today's forest plot despite being the day's largest study.
- **One directly actionable siting result.** Ianiri et al. (*Environ Pollut*) find street-level deposition fluxes roughly **1.6× roof-level** (42 vs 26 $mg\,m^{-2}\,d^{-1}$ annual SPM) but *no* significant street–roof difference once pollutant loads are normalised to collected SPM (Wilcoxon, $p > 0.05$). Sampling height changes how much dust you collect, not what the dust is made of — which is the assumption most rooftop-versus-breathing-zone network designs implicitly make and rarely test.
- **Three optical-property records in one day, and two ship without an abstract.** The Halo Doppler lidar depolarisation work is complete and quantitative; the AS&T humidified CAPS-PM_{SSA} paper and the *J Aerosol Sci* refractive-index inversion are carried on title and author only, because Taylor & Francis and Elsevier both withheld the abstract and Semantic Scholar was rate-limited for the whole run. Both are labelled `Metadata only` rather than pooled with characterised designs. The nine rejections, including two scope calls on PM-adjacent reviews, are itemised in the provenance note.

Corpus analytics

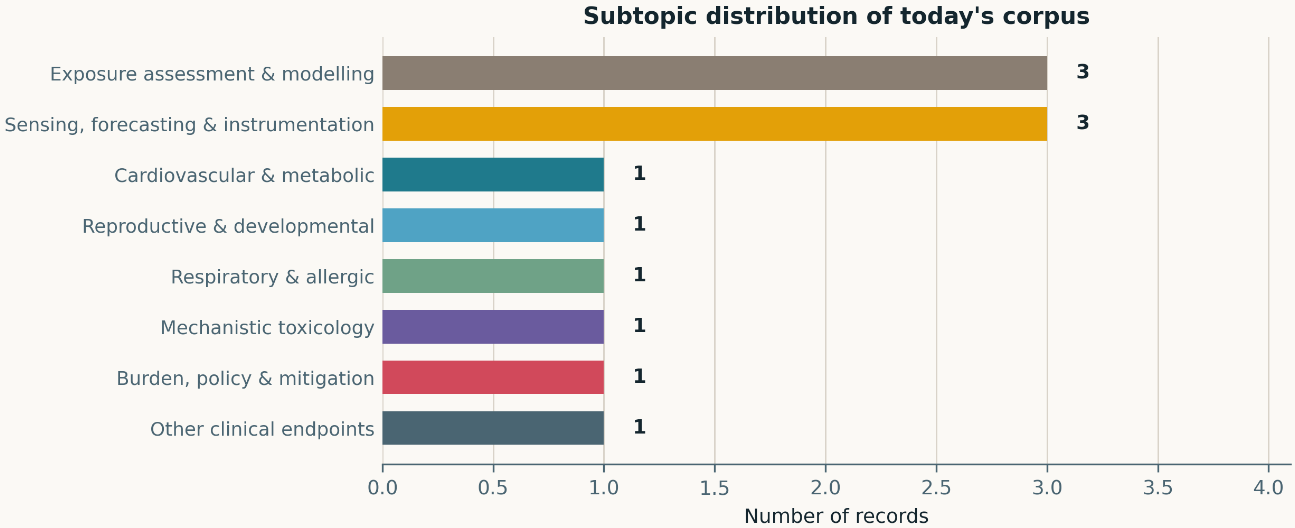
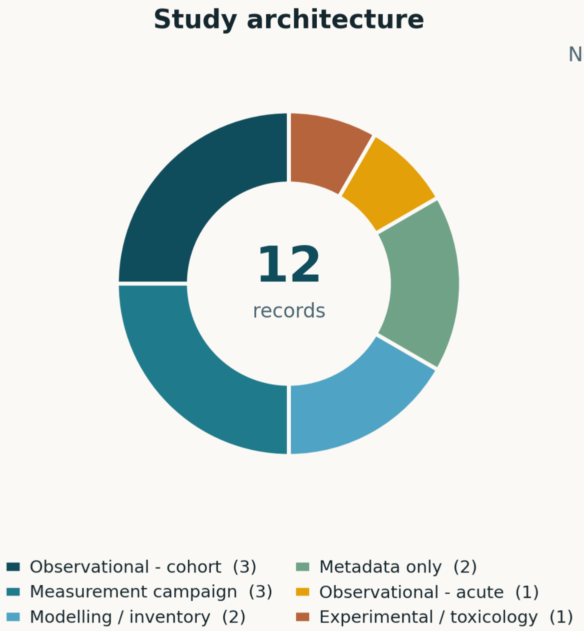


Fig. Subtopic assignment of the 12 carried records, single-label by dominant contribution. Eight of ten clusters are occupied and no cluster holds more than three, which is 0.67 clusters per record — **the most dispersed issue of the watch to date**, ahead of 14 August (0.64) and 31 July (0.64), computed over all 23 issues. *Sensing, forecasting & instrumentation* and *Exposure assessment & modelling* tie at 3 each; six clusters hold exactly one record. *Neuro / mental health* is empty for the twelfth time in twenty-three issues and *Occupational & indoor* for the sixth, both counts computed from `state/metrics.csv` rather than recalled. Twelve records ties 9 August as the fourth-smallest issue of the watch, behind 10 (5 and 16 August) and 11 (31 July).



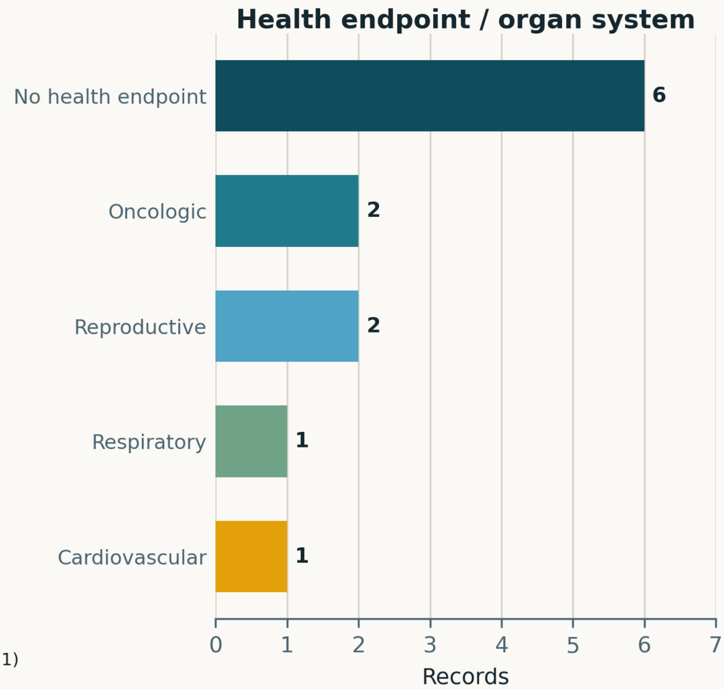


Fig. Left — study architecture. Observational cohorts lead at 3 (Nanjing pregnancies, Korean leukaemia, UK Biobank proteomics), matched by 3 measurement campaigns (school metals, Rome deposition, Halo lidar). ***Metadata only* takes 2 of 12** and is shown as its own slice by design: pooling an uncharacterised record with a characterised one is how the architecture panel was quietly wrong before this category existed. Right — health endpoint. **No health endpoint** is again the largest bar at 6 of 12 (50%), sitting mid-band in the 41–67% range of the previous ten issues (7–16 August: 54, 64, 67, 56, 58, 46, 41, 43, 43, 60%).

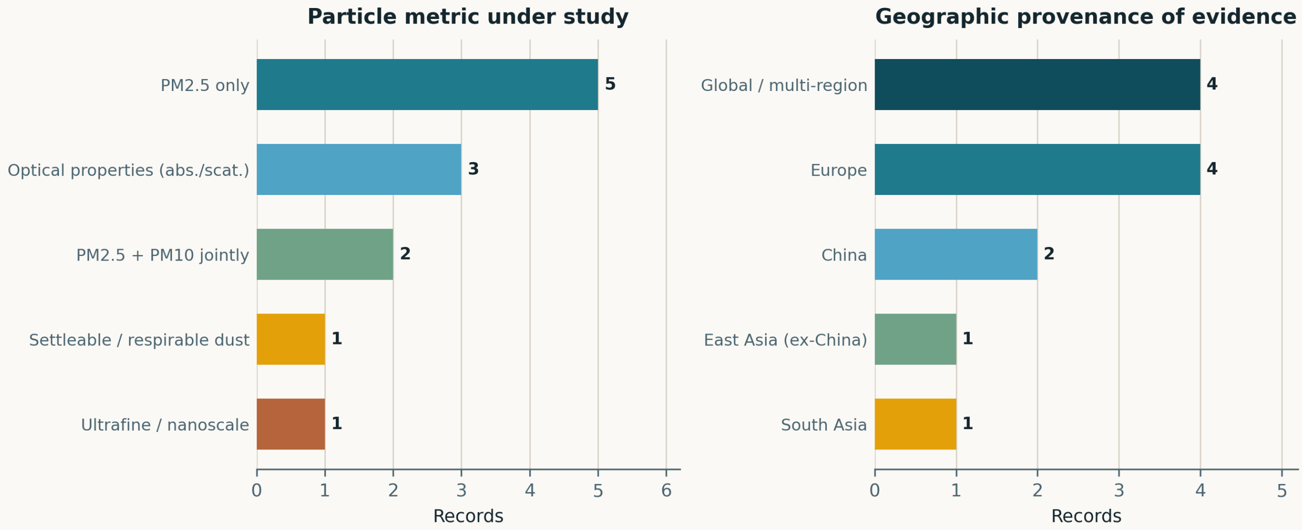


Fig. Left — particle metric. $PM_{2.5}$-only accounts for 5 records and $PM_{2.5}$ with PM_{10} jointly for 2; **optical properties (absorption/scattering) take 3**, the whole instrumentation block, and one record each covers ultrafine carbon black and settleable dust. Right — geography. Europe leads at 4 (Sweden, Italy, Finland, UK). **The *Global / multi-region* slice of 4 should be read as an absence, not as reach**: only the GLENS preprint is genuinely global; the other three are a murine laboratory study, a bench instrument paper and a metadata-only algorithm record with no stated field site. South Asia's single record is the Pakistan satellite study, the first since 15 August; it is also the only record today from a region carrying a top-decile attributable burden.

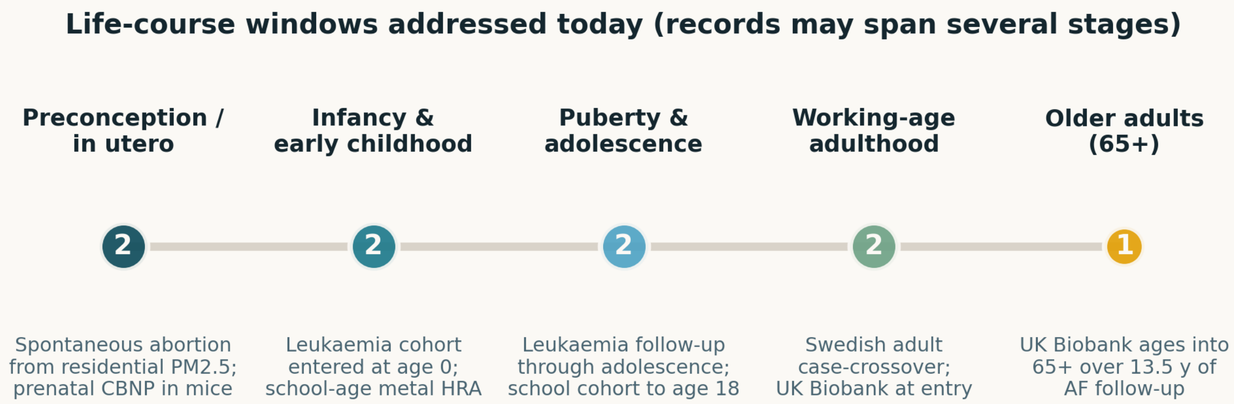


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. Preconception/in-utero, infancy, adolescence and working age each hold 2; the 65+ window holds 1 and only by inference, since UK Biobank participants age into it across 13.5 years of follow-up rather than being recruited there. The leukaemia cohort contributes to three consecutive windows because it enrols at birth and follows past age 18, which is the design feature that distinguishes it from the rest of the paediatric literature.

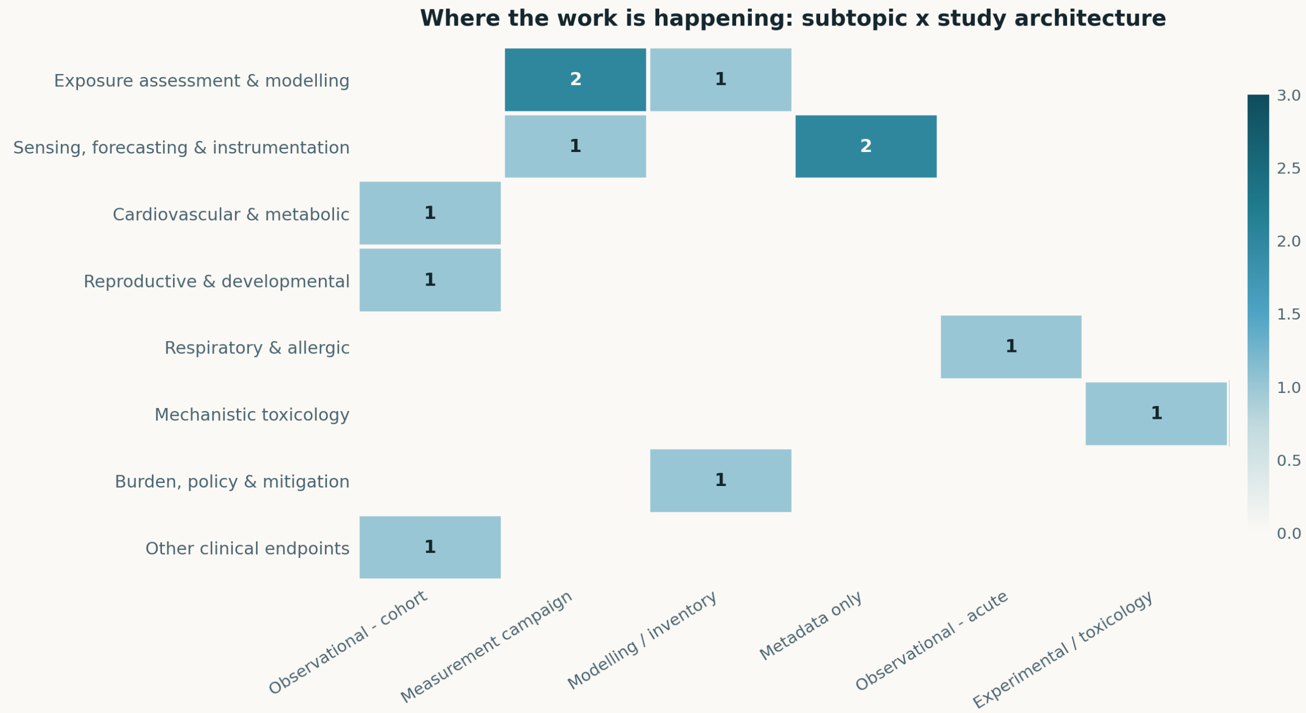


Fig. Subtopic × study-architecture cross-tabulation. Twelve records occupy 9 of the 80 available cells and no cell holds more than 2. The silo persists in the same shape as the last three issues: every *Sensing* and *Exposure* record sits in a measurement, modelling or metadata-only column and every human-endpoint record in an observational one, with no cell bridging them. Note that the two *Metadata only* entries sit entirely inside *Sensing* — the publishers that withhold abstracts on this watch are the aerosol-instrumentation publishers, so the gap is systematic rather than incidental.

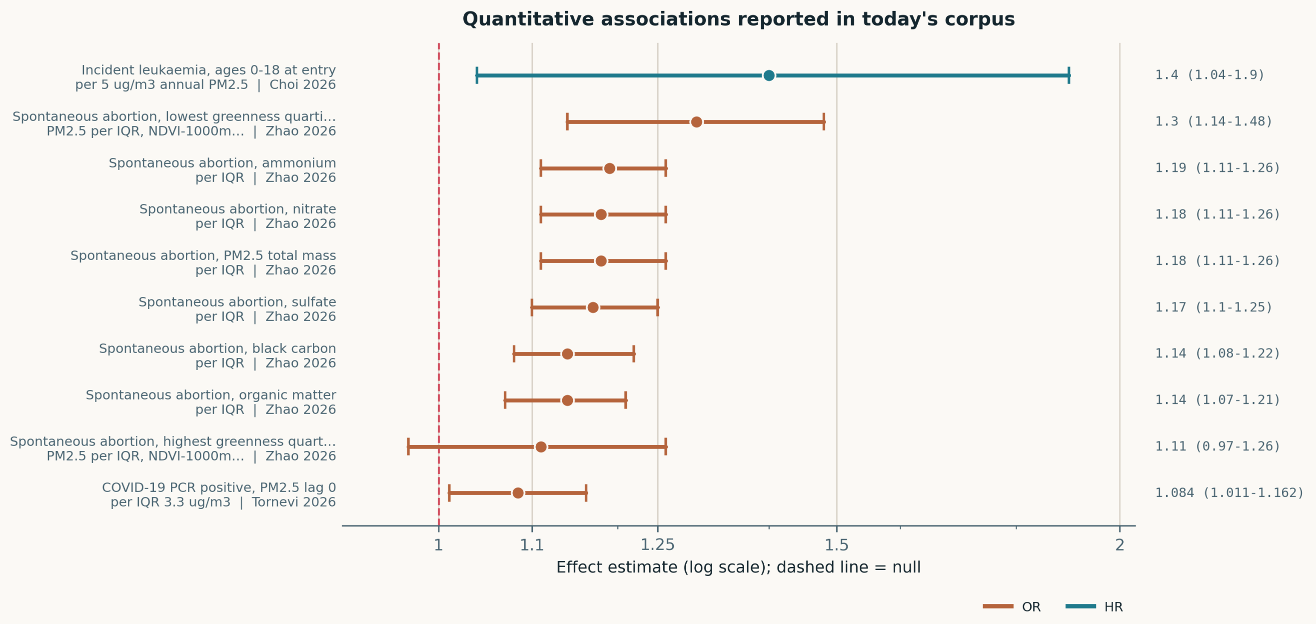


Fig. The 10 interval estimates carried today, on a common logarithmic axis. **Eight of the ten come from one study and are not independent.** The six Zhao constituent estimates are IQR-scaled contrasts on correlated fractions of the same aerosol in the same 20,117 pregnancies, so their *ordering* is informative and their spread is not a between-study spread; the two greenness strata are subsets of that same cohort and share person-time with all six. Only Tornevi (per IQR $3.3\,\mu g/m^3$, 1-day lag, case-crossover) and Choi (per $5\,\mu g/m^3$, annual mean, cohort) are independent of it and of each other, and their exposure contrasts differ by roughly an order of magnitude in both averaging time and increment. Read within family, never across.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Cardiovascular & metabolic 1 record

Chen et al. 2026 Ecotoxicol Environ Saf

Pollutant-specific plasma proteomic signatures and atrial-fibrillation-free life lost

UK Biobank, ~2,900 Olink plasma proteins screened against long-term $PM_{2.5}$, PM_{10}, NO_2, NO_x, SO_2, benzene and O_3 under Bonferroni correction, then elastic-net regularisation to build one signature per pollutant; incident AF over a median **13.53 y** by Cox regression. Signatures out-predicted the external exposure metrics and stratified risk after adjusting for ambient concentration; high-risk participants lost an estimated **1.15–1.25 AF-free years**. High-throughput mediation nominates **ANGPT2, GDF15 and PLAUR**. The weakness is structural, not analytic: a signature trained on a modelled exposure inherits that model's error and its spatial confounding, so "out-performs the exposure metric" may mean "encodes residence better". No HR or interval is reported, so nothing enters today's forest plot.

doi: 10.1016/j.ecoenv.2026.120666

Reproductive & developmental 1 record

Zhao et al. 2026 Chin Med J (Engl)

$PM_{2.5}$ constituents and spontaneous abortion, modified by residential greenness

20,117 pregnancies in Nanjing with residential $PM_{2.5}$ speciated into NO_3^-, SO_4^{2-}, NH_4^+, BC and OM; logistic regression with propensity-score matching plus two-pollutant models, WQS and BKMR for the joint effect. Per IQR: total mass **OR 1.18 (1.11–1.26)**, ammonium **1.19 (1.11–1.26)** highest, BC and OM lowest at 1.14 — and OM and BC alone lose significance once season of conception is added. Greenness (NDVI-1000 m) interacts significantly but non-monotonically (Q1 1.30, Q2 1.02, Q3 1.29, Q4 1.11), which undercuts the attenuation reading. Depression and anxiety scores predicted SAB independently with no interaction. Constituent correlations are high, so the ordering is suggestive rather than separable.

doi: 10.1097/CM9.0000000000004259

Respiratory & allergic 1 record

Tornevi et al. 2026 BMJ Open Respir Res

Short-term urban-background PM and the timing of confirmed COVID-19, three Swedish cities

Case-crossover on **3,999** PCR-confirmed cases from a prospective adult cohort in Stockholm, Gothenburg and Uppsala, 7 Mar 2020 – 26 Jan 2022, with daily $PM_{2.5}$, PM_{10} and NO_2 from urban-background stations and distributed-lag (and non-linear) models adjusted for meteorology. **$PM_{2.5}$ at lag 0: OR 1.084 (1.011–1.162) per IQR 3.3 $\mu g/m^3$**, consistent across all three cities; PM_{10} most consistent at lag 3; NO_2 lag 0 only after dropping the final wave. The authors themselves read the same-day lag as implausible for infection and propose symptom exacerbation prompting testing — i.e. the outcome is a *test date*, not an infection date, and the estimate is at least partly a health-seeking-behaviour effect.

doi: 10.1136/bmjresp-2025-003279

Mechanistic toxicology 1 record

Li et al. 2026 FASEB J

Size-resolved carbon-black nanoparticle effects on the maternal–fetal lung axis

Pregnant mice exposed to 30 nm or 120 nm carbon black under an explicitly high-dose mechanistic protocol, with H&E/Masson histology, dark-field hyperspectral imaging for particle localisation and chemical-isotope-labelling LC-MS metabolomics. Both sizes produced maternal inflammatory infiltration and fibrosis, **120 nm worse than 30 nm**; transplacental transfer was imaged directly in fetal lung. Fetal lungs showed no histological injury but raised oxidative and inflammatory markers, with size-specific metabolic signatures — 30 nm on histidine metabolism, 120 nm on butanoate and arachidonic acid. Two problems: the dose is not environmentally referenced, and the larger particle causing more damage inverts the surface-area expectation without a number- versus mass-dose accounting that would explain it.

doi: 10.1096/fj.202600065RR

Exposure assessment & modelling 3 records

Moryani et al. 2026 J Appl Toxicol

$PM_{2.5}$-bound Cu, Cd, Ni, Pb and Zn across 20 schools in two contrasting South China cities

Paired summer and winter sampling at 20 schools in Guangzhou and Maoming, ICP-MS for five metals, with geoaccumulation index, enrichment factor and PCA source apportionment, then inhalation risk assessment for children and adults. Cu and Cd carried the highest enrichment (EF 27.72 and 27.32 in Guangzhou), winter exceeded summer, and PCA pointed to traffic, industrial combustion and urban emissions. **All hazard quotients and carcinogenic risks fell below accepted thresholds**, with Pb (4.85×10^{-3}) and Cu (3.27×10^{-3}) leading non-carcinogenic risk and Cd (7.30×10^{-6}) carcinogenic. Fieldwork dates from 2018, so it characterises an emissions regime seven years old; the null risk result is the honest headline and is under-stated in the abstract.

doi: 10.1002/jat.70398

Ianiri et al. 2026 Environ Pollut

Street- versus roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM

Thirteen monthly deposimeter samples at two heights on one Rome building, Aug 2023 – Aug 2024, with PCDD/Fs, PCBs and PAHs by GC-HRMS and metals by ICP-MS, reported both as fluxes and normalised to collected sedimentable PM. **Annual SPM flux 42 vs 26 mg m^{-2} d^{-1}** street versus roof, but a Wilcoxon signed-rank test found **no significant street–roof difference in SPM-normalised concentrations** ($p > 0.05$): height changes deposited quantity, not deposited composition. Seasonality was significant for PCDD/Fs, PCBs, Cr, As, Ni and Sb (winter peaks) and for PAHs (summer/autumn). Single building, single city, $n = 13$ per height — a design that supports the composition claim far better than the flux ratio.

doi: 10.1016/j.envpol.2026.128965

Awan et al. 2026 Aerosol Air Qual Res

BRIQ-PM$_{2.5}$: quantile-mapping attribution of industrial excess PM$_{2.5}$ across Pakistan

Proposes estimating industrial impact as the residual between an observed satellite PM$_{2.5}$ distribution and the same distribution quantile-mapped onto a province-specific "low-industry" baseline city, then interpolating the residual by variogram modelling and universal kriging across four provinces. Reported validation is distributional — histogram and ECDF alignment, improved RMSE after correction. **The attribution is definitional, not estimated**: whatever separates a target city from its chosen baseline is labelled industrial, so meteorology, dust, biomass burning and population density all load onto the same term, and no ground-monitor comparison or sensitivity analysis over baseline choice is reported. Useful as a hotspot screen; not an emissions attribution.

doi: 10.1007/s44408-026-00160-z

Sensing, forecasting & instrumentation — 3 records

Filioglou et al. 2026 Atmos Chem Phys

Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar

Assesses whether a wind-profiling Halo Photonics StreamLine Doppler lidar can type aerosol at 1565 nm, a wavelength where depolarisation is rarely characterised, across three tropospheric case studies. **Extremely fresh smoke gave 0.017 $\pm$ 0.004**, aged long-range-transported smoke marginally higher, and **volcanic ash layers 0.45 $\pm$ 0.01** — a separation wide enough to type layers unambiguously. Smoke mass concentration retrieved from the 1565 nm signal agreed with concurrent in-situ observations. The value here is that an instrument already deployed for wind now also delivers aerosol typing and a mass estimate on the same beam; three case studies is thin for a conversion factor, and the aged-smoke result is qualitative.

doi: 10.5194/acp-26-11525-2026

Dal Porto et al. 2026 Aerosol Sci Technol

Humidified CAPS-PM$_{\mathrm{SSA}}$ with a revised truncation-correction methodology

Metadata only. Taylor & Francis withheld the abstract on this run and Semantic Scholar was rate-limited throughout, so nothing below the title, authors and journal is verified and no result is asserted. On the registered title alone the paper describes a humidified variant of the cavity-attenuated phase-shift single-scattering-albedo monitor together with an improved method for the scattering truncation correction — the two quantities that dominate uncertainty when an SSA measurement is used as a reference for hygroscopic growth. Relevant enough to the humidity-correction problem to carry rather than defer; flagged for abstract retrieval on the next run.

doi: 10.1080/02786826.2026.2711663

Zhao et al. 2026 J Aerosol Sci

Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar

Metadata only. Elsevier returned no abstract through Crossref and ScienceDirect did not serve the record; no finding is asserted. The registered title describes an improved Monte-Carlo sampling algorithm that inverts the aerosol complex refractive index from multi-wavelength lidar signals combined with an independently measured particle size distribution. Carried at tier C because refractive index is the parameter that converts an optical measurement into a mass estimate, and every low-cost nephelometric sensor assumes one implicitly; whether this inversion is well-posed cannot be judged from the title.

doi: 10.1016/j.jaerosci.2026.106883

Burden, policy & mitigation — 1 record

Xia et al. 2026 (preprint) arXiv

Surface PM$_{2.5}$ mass and composition under stratospheric sulfate geoengineering

Uses the GLENS ensemble (CESM1, stratospheric SO_2 injection balancing RCP8.5 forcing) to ask what solar geoengineering does to surface PM$_{2.5}$. Global mean PM$_{2.5}$ *falls* relative to RCP8.5, but the reduction is carried entirely by **less dust** (wetter soils, higher leaf-area index over deserts) and **less sea salt** (slower winds). Excluding those two, PM$_{2.5}$ *rises*, driven by aerosol-phase secondary organic aerosol partitioning in a cooler atmosphere. Notably, the injected SO_2 does **not** raise surface sulfate. Single model, single ensemble, an RCP8.5 baseline now far from plausible, and SOA gas precursors are prescribed — so the SOA increase is partly an artefact of that setup, as the authors acknowledge. Preprint; not peer reviewed.

doi: 10.48550/arXiv.2608.14928

Other clinical endpoints — 1 record

Choi et al. 2026 Environ Res

Time-varying long-term PM$_{2.5}$ and leukaemia from childhood into early adulthood

National Health Insurance Service–National Sample Cohort, 2002–2020: **384,606** people aged 0–18 at enrolment, **272 incident leukaemias** (83 diagnosed after age 18), annual PM$_{2.5}$ assigned to all 226 inland districts from an ML ensemble, Cox models with time-varying exposure and confounders plus two-pollutant sensitivity against NO_2 and O_3. **HR 1.40 (1.04–1.90) per 5 μg/m^3**, stable when censored at 18. The exposure contrast is the problem: cohort mean **29.6 $\pm$ 3.1 μg/m^3**, so a 5 μg/m^3 increment exceeds 1.5 SD of the between-district spread and the estimate extrapolates beyond most of its own support, on 272 events with district-level assignment and no residential-mobility handling.

doi: 10.1016/j.envres.2026.125497

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Chen et al.	Pollutant-specific plasma proteomic signatures, mediators and atrial-fibrillation-free life lost	*Ecotoxicol Environ Saf*	Prospective cohort	Long-term modelled $PM_{2.5}$ and PM_{10} (plus NO2, NOx, SO2, benzene, O3)	UK
Reproductive & developmental					
Zhao et al.	$PM_{2.5}$ constituents and spontaneous abortion, modified by residential greenness	*Chin Med J (Engl)*	Prospective cohort (mixtures)	Residential annual $PM_{2.5}$ with NO3-, SO42-, NH4+, BC and OM fractions	China
Respiratory & allergic					
Tornevi et al.	Short-term urban-background $PM_{2.5}$ and the timing of confirmed COVID-19, three Swedish cities	*BMJ Open Respir Res*	Case-crossover	Daily urban-background $PM_{2.5}$ and PM_{10} from fixed reference monitors	Sweden
Mechanistic toxicology					
Li et al.	Size-resolved carbon-black nanoparticle effects on the maternal-fetal lung axis	*FASEB J*	Animal model	Ultrafine carbon-black nanoparticles, 30 nm and 120 nm, high-dose mechanistic exposure	Laboratory (murine)
Sensing, forecasting & instrumentation					
Dal Porto et al.	Humidified CAPS-PM_SSA with a revised truncation-correction methodology	*Aerosol Sci Technol*	Metadata only (no abstract)	Aerosol optical extinction and scattering, single-scattering albedo	Chamber / laboratory
Filioglou et al.	Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar: smoke vs volcanic ash	*Atmos Chem Phys*	Multi-instrument field evaluation	Lidar depolarisation ratio and aerosol layer optical properties at 1565 nm	Finland
Zhao et al.	Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar plus size distribution	*J Aerosol Sci*	Metadata only (no abstract)	Complex refractive index, multi-wavelength lidar backscatter, particle size distribution	Not stated
Exposure assessment & modelling					
Awan et al.	BRIQ-$PM_{2.5}$: quantile-mapping attribution of industrial excess $PM_{2.5}$ across Pakistan	*Aerosol Air Qual Res*	Spatial analysis / GIS	Satellite-derived surface $PM_{2.5}$	Pakistan
Ianiri et al.	Street- vs roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM	*Environ Pollut*	Deposition sampling	Settleable particulate matter, monthly deposimeter collection	Italy
Moryani et al.	$PM_{2.5}$-bound Cu, Cd, Ni, Pb and Zn in 20 schools across two contrasting South China cities	*J Appl Toxicol*	Source apportionment	Heavy-metal content of $PM_{2.5}$, summer and winter	China
Burden, policy & mitigation					
Xia et al. 2026	Surface $PM_{2.5}$ mass and composition under stratospheric sulfate geoengineering	*arXiv* (*preprint*)	Chemical transport modelling	Modelled surface $PM_{2.5}$ mass and speciation (dust, sea salt, OC, sulfate, SOA)	Global
Other clinical endpoints					
Choi et al.	Time-varying long-term $PM_{2.5}$ and leukaemia incidence from childhood into early adulthood	*Environ Res*	Nationwide cohort	Annual district $PM_{2.5}$ from a machine-learning ensemble, 226 inland districts	South Korea

Provenance and method

Entry window **2026-08-17 to 2026-08-17**, contiguous with the 16 August issue; no gap. **Per-source counts.** PubMed connector on two separate axes returned **9** on the health axis and **1** on the sensing axis (the sensing hit was already in the health set); the local `pubmed_health` leg returned 7 and `pubmed_sensing` 1, all duplicates. Europe PMC on the primary particle query returned **1**, and a broader air-quality sweep **6** of which 2 were new and in scope. Crossref-by-ISSN returned **9 across 18 journals** and carried the whole instrumentation block — *Atmos Chem Phys*, *Aerosol Sci Technol*, *J Aerosol Sci* and *Aerosol Air Qual Res* are not indexed in PubMed, so without this leg today's issue would have had no sensing records at all. arXiv contributed **1** under a 3-day recency screen (a 14 August posting the 15 and 16 August sweeps both missed, which is the second time the relevance query's unstable ordering has been caught by that screen). **OpenAlex HTTP 429, 15th consecutive** — the leg is dead and is no longer retried. ClinicalTrials.gov returned **0** PM or air-pollution registrations created or updated in the window; `trials.json` unchanged.

Two records are metadata-only. Taylor & Francis and Elsevier both withheld the abstract for `10.1080/02786826.2026.2711663` and `10.1016/j.jaerosci.2026.106883`, and the Semantic Scholar fallback was rate-limited for the duration of the run. Both are labelled `Metadata only (no abstract)` in the register, given their own slice in the architecture panel, and assert no finding beyond their registered titles. Both are flagged for abstract retrieval on the next run. One record is a **preprint** (Xia et al., arXiv) and is labelled as such.

Rejections (9). Two out-of-scope PubMed hits (a black-carbon crop-physiology review with no human endpoint or monitoring method; a family-medicine climate review naming air pollution only as background); two Europe PMC drug-delivery records that matched on `aerosol`/deposition terms but concern intranasal powder dosimetry and inhalable antibacterial nanosystems; and five Crossref-by-ISSN records from in-scope journals that are out of scope by subject — aerosol–cloud effects on convective precipitation, urban heat islands, Southern Ocean methylated sulfur, airborne eddy-covariance VOC fluxes, and an opinion piece on AI in aerosol science. All logged with reasons to `state/rejected.jsonl`.

Labels. Three geography keys were added to `plots.py` this issue (`Sweden`, `Finland`, `Pakistan`) plus four Swedish city needles; `Sweden` had neither a key nor a needle and would have fallen to the unmapped bucket, and `Pakistan` resolved to the correct group only by routing through an `India` needle. No other new leaf label was written and the unmapped-label diagnostics printed nothing. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Every claim of the form "first/most/widest since X" in the captions above was computed from `state/metrics.csv`, not recalled. `check_dois.py` returned **0 fail, 4 warn**; three warnings are title-overlap heuristics firing on editorial focus lines and one is arXiv's absence from Crossref, each confirmed by eye against the registered record.